

Multiple Choice

1. **Answer:** B

Options: A) CH₃F; B) CH₃Cl; C) CH₃Br; D) CH₃I

Note: Correct option: B - CH₃Cl

2. **Answer:** D

Options: A) Q = 1012; B) Q = 2012; C) Q = 3012; D) Q = 6012

Note: Correct option: D - Q = 6012

3. **Answer:** D

Options: A) Strong oxidizing conditions; B) Low-spin electron configuration; C) Metallic physical properties; D) Unpaired electrons

Note: Correct option: D - Unpaired electrons

4. **Answer:** A

Options: A) 17 O; B) 19 F; C) 29 Si; D) 31 P

Note: Correct option: A - 17 O

5. **Answer:** A

Options: A) Neutrons; B) Alpha particles; C) Beta particles; D) X-rays

Note: Correct option: A - Neutrons

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The total entropy of the system plus surroundings increases.'

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Both ³⁵Cl and ³⁷Cl have electric quadrupole moments.'

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Ultraviolet. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key